3 Carlo puts tins into small boxes and into large boxes.

He puts 6 tins into each small box.
He puts 20 tins into each large box.

Carlo puts a total of 3000 tins into the boxes so that

number of tins in small boxes : number of tins in large boxes = 2 : 3

Carlo says that less than 30% of the boxes filled with tins are large boxes.

Is Carlo correct?
You must show all your working.

3000 ÷ (2 + 3) = 600 tins per ratio part.

Small-box tins = 2 × 600 = 1200; large-box tins = 3 × 600 = 1800.

Small boxes = 1200 ÷ 6 = 200; large boxes = 1800 ÷ 20 = 90.

Large-box share = 90/(200 + 90) × 100 ≈ 31.0%.

No: 31.0% is greater than 30%.

5 marks total
600 per part
1200 / 1800 tins
200 / 90 boxes
90/290 ≈ 31%
No, supported

(Total for Question 3 is 5 marks)

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